

Project Summary: 2 Stage Reduction Gearbox Analysis

Reduction Gearbox · Personal Engineering Project

① Inputs & Requirements

Inputs — FS Final-Drive Gearbox

CBR600RR Engine Outputs from Transmission	
Input speed, n_{in}	11500 rpm
Input torque, T_{in}	63.0 N·m
Ratio (fixed tooth counts, per spec)	
Stage 1 ratio, i_1 ($z_1=20, z_2=40$)	1.58
Stage 2 ratio, i_2 ($z_1=24, z_2=48$)	1.58
Overall ratio achieved, $i_1 \times i_2$	2.50
Efficiency & Resulting Output	
Assumed efficiency per stage	95.0%
Overall efficiency (2 stages)	90.3%
Output speed	4606.6 rpm
Output torque	141.9 N·m
Manufacturing & Mounting (confirmed)	
ISO 1328 quality grade	6
AGMA-equivalent quality number, $Q_v = 1$	6.0
Application factor, K_A (as specified)	1.50
Design life target	20 hours
Gear Geometry Common Inputs	
Normal pressure angle, α_n	20 deg
Helix Angle N/A for Spur	
Material Gears - 16 Mn Cr 5	
Bending endurance limit, σ_{Flim} (IS)	430 MPa
Contact endurance limit, σ_{Hlim} (IS)	1500 MPa
Material Shaft - 42CrMo4	
Ultimate Tensile Strength S_{Ut}	900 MPa
Yield Strength S_y	680 MPa
Minimum Value	680 MPa

Inputs derived from Honda CBR600RR engine specification. Materials were chosen from common choices made in industry.

② Analysis

K_v was calculated using AGMA 2001:D04 equations. This replaces a graphical method used in ISO 2019. Grade 6 quality was chosen to reflect the conditions the gearbox undergoes.

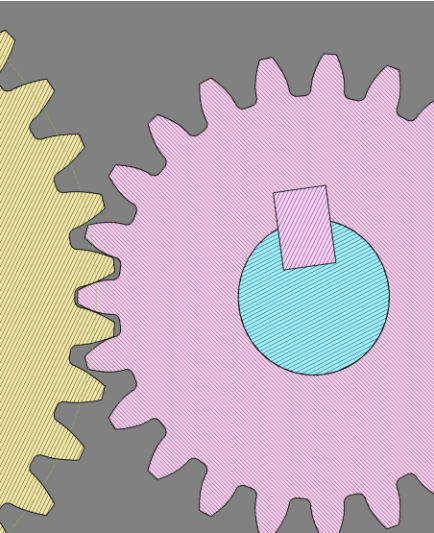
Tooth Counts & Ratio Check	
Pinion tooth count, z_1	23
Gear tooth count, z_2	36
Gear ratio, $u = z_2/z_1$	1.57
Target stage ratio (from Inputs)	1.58
Geometry	
Normal module, m_n	2.50 mm
Face width, b	25.0 mm
Pinion pitch diameter	57.50 mm
Gear pitch diameter	90.00 mm
Center distance,	73.75 mm
Face width ratio	0.43
Loads	
Input speed at pinion	11500.0 rpm
Input torque at pinion	63.0 N·m
Pitch line velocity	34.62 m/s
Force Tangent F_t	2191.3 N
Contact Ratio Calculations	
Pinion base radius,	27.02
Gear base radius	42.29
Pinion tip radius	31.25
Gear tip radius	47.50
Contact Ratio	1.642

Geometry Calculations for Gear stage 1: 23/36.

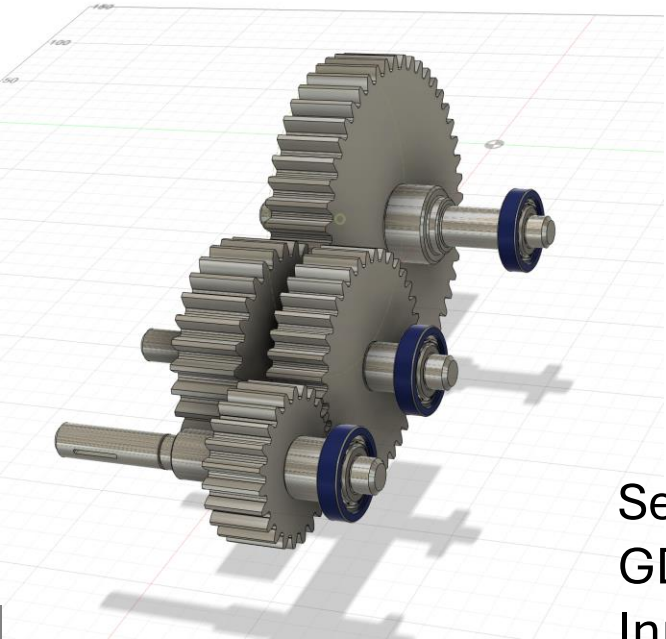
Calculations involving ISO 6336 load factors (K , Y and Z factors) are not reproduced here; values were obtained from the referenced standard and applied within the calculation.

Top-Level Performance	
Overall ratio	2.50
Overall efficiency	90.3%
Output torque (N·m)	141.9
Output speed (rpm)	4606.6
Gear Safety Factors (ISO 6336)	
Stage 1 — bending SF	2.10
Stage 1 — contact SH	1.27
Stage 1 — pitch line velocity (m/s)	34.62
Stage 1 — K_v (dynamic factor)	2.05
Stage 2 — bending SF	2.83
Stage 2 — contact SH	1.46
Stage 2 — pitch line velocity (m/s)	28.58
Stage 2 — K_v (dynamic factor)	1.96
Shaft Minimum Diameters (mm)	
Shaft 1 — Input (from CBR primary reduction)	15.21
Shaft 2 — Intermediate	17.41
Shaft 3 — Output (to differential)	19.94

Overall FoS reached was 1.27, increasing module, face width or tooth count will increase this. Minimum Shaft diameters were calculated and used to select values from standard diameters. (15, 17, 20, 25, 30)



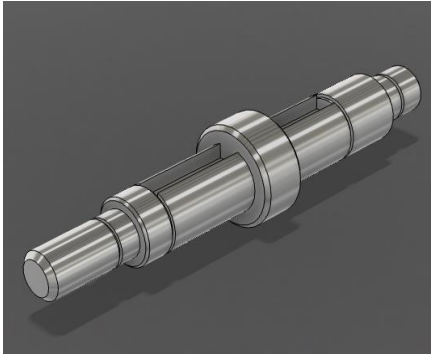
③ Assembly and Drawings



Assembly of Bearings, Shafts and Gears in Autodesk Fusion 360. *Housing Not Shown* Revolute Joints and Motion Links used to showcase gear motion.

See next page for GD&T drawings of Input Shaft

Gears and Shaft power transfer by Keyway cutout and insert. Section analysis shows gear mesh of Stage 1



Shown Intermediate Shaft. Shaft diameters from calculation of 20mm. Min 17.41mm calculated using ISO 6336.

Circlip cutouts included to retain bearings and gears in place.

